AARAV DESHMUKH
UAV Airframe & Composites Intern | SOLIDWORKS | CFRP | FEA | PX4 | UAV Airframe Design
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https://linkedin.com/in/aarav-demo | https://github.com/aarav-demo/engineering-portfolio

SUMMARY

Mechanical engineering graduate with dated internship and competition hardware experience. Airframes, composites & flight systems.

EDUCATION

Pune Institute of Technology | B.Tech in Mechanical Engineering | Jun 2026
CGPA: 8.6 / 10.0

INTERNSHIPS

Aero Works - UAV Airframe Intern | Bengaluru, Karnataka | Jan 2026 - Jul 2026
- Designed a UAV airframe in SOLIDWORKS with Carbon Fiber Reinforced Polymer (CFRP) composite layup and vacuum bagging, reducing airframe mass by 18%.
- Validated landing gear impact using Finite Element Analysis (FEA) in ANSYS Mechanical, achieving a Factor of Safety (FOS) of 1.9 under 4g load.
- Integrated a PX4 and ArduPilot Pixhawk controller using MAVLink into a Vertical Take-Off and Landing (VTOL) airframe, increasing flight endurance by 12%.

ENGINEERING PROJECTS

Flight Competition Team - Airframe Lead | Pune, Maharashtra | Aug 2025 - Dec 2025
- Optimized a Brushless DC Motor (BLDC) propeller assembly with ANSYS Fluent and XFLR5, increasing thrust-to-weight by 15%.
- Fabricated a landing gear fixture in SOLIDWORKS and documented DGCA type certification test evidence, achieving 0.1 mm tolerance.

TECHNICAL SKILLS

SOLIDWORKS | Carbon Fiber Reinforced Polymer (CFRP) | Finite Element Analysis (FEA) | PX4 | UAV Airframe Design

CERTIFICATIONS

Engineering design and hardware safety workshop | Feb 2026